INF70005 STRATEGIC PROJECT MANAGEMENT

Project selection & prioritisation

Contents

- Process Groups / Life Cycle
- Ideas!
- Project selection and benefits management
- Methods for testing feasibility
- CSF’s & KPI’s

Project Process Groups / Life cycle

and PMBoK Knowledge areas …

Project Process Groups

- Initiate
- Planning
- Execute
- Monitoring / Controlling
- Closing

Project Management Body of Knowledge (PMBoK)

Originally developed by Project Management Institute (USA) in 1984; current is 7th edition 2021.

PMBoK *6th edition was based more on processes, tools and techniques, the 7th edition has an over-arching focus principles.*

The PMBoK does not provide all the answers, but is a useful guide if used wisely …

PMBOK® Process Groups

Initiating—“define a project or a new phase by obtaining authorization”

Planning—“establish the project scope, refine objectives and define actions to attain objectives”

Executing—“complete the work defined to satisfy project specifications”

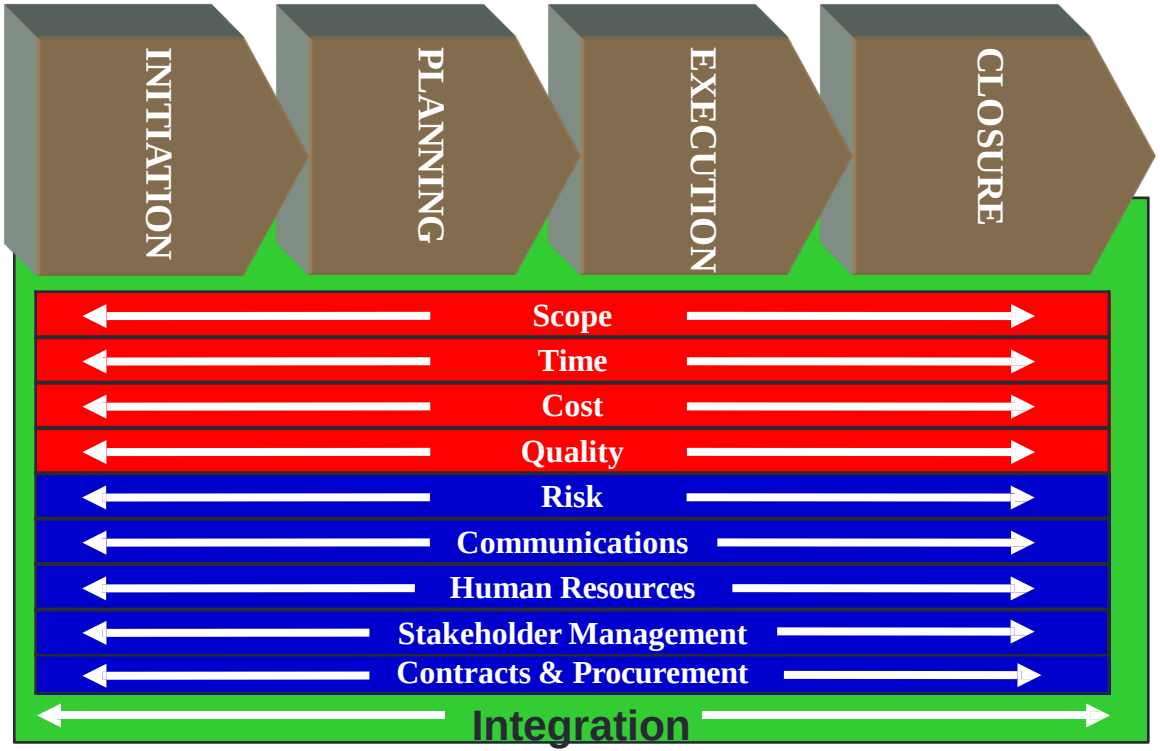
Monitoring and controlling—“track, review, and regulate progress and performance, identify changes required, and initiate changes”

Closing—“finalize all activities to formally to close project”

Project Process Groups

- Process Groups cover the full Life Cycle of the Project
- Each Process Group (Phase) makes a clear contribution to the resolution of the business problem the project is intended to address

PMBoK knowledge areas



PMBOK® Knowledge Areas

Integration management - “processes and activities to identify, define, combine, unify, and coordinate the various processes and project management activities”

Scope management - “processes to ensure that the project includes all the work required, and only the work required, to complete the project successfully”

Time management - “processes to manage timely completion of the project”

PMBOK® Knowledge Areas

Cost management – "processes involved in planning, estimating, budgeting, financing, funding, managing, and controlling costs so that the project can be completed within the approved budget"

Quality management - "processes and activities of the performing organization that determine quality policies, objectives, and responsibilities so that the project will satisfy the needs for which it was undertaken"

Human Resources management - "processes that organize, manage, and lead the project team"

PMBOK® Knowledge Areas

Communications management - “processes to ensure timely and appropriate planning, collection, creation, distribution, storage, retrieval, management, control, monitoring, and ultimate disposition of project information”

Risk management - “processes of conducting risk management planning, identification, analysis, response planning, and control…to increase the likelihood and impact of positive events and decrease the likelihood and impact of negative events in the project”

PMBOK® Knowledge Areas

Stakeholder management - “processes to identify the people, groups, or organizations, that could impact or be impacted by the project, analyze their expectations and impact, and develop strategies for engaging them and managing conflicting interests”

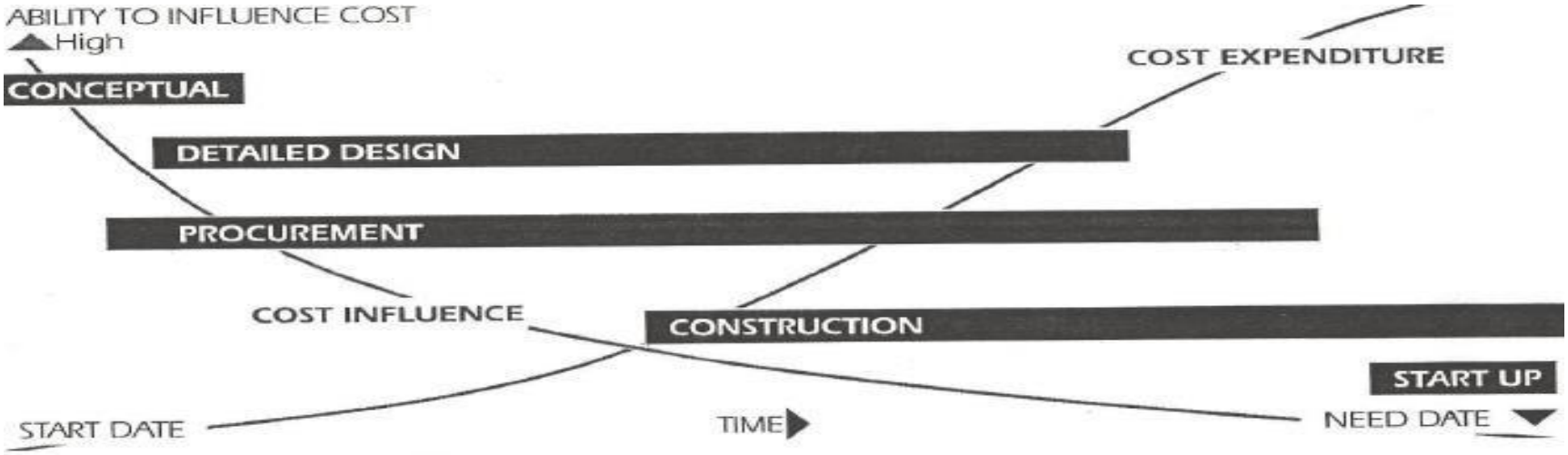
Procurement management - “processes to purchase or acquire products, services, or results from outside the project team”

Project Initiation Phase

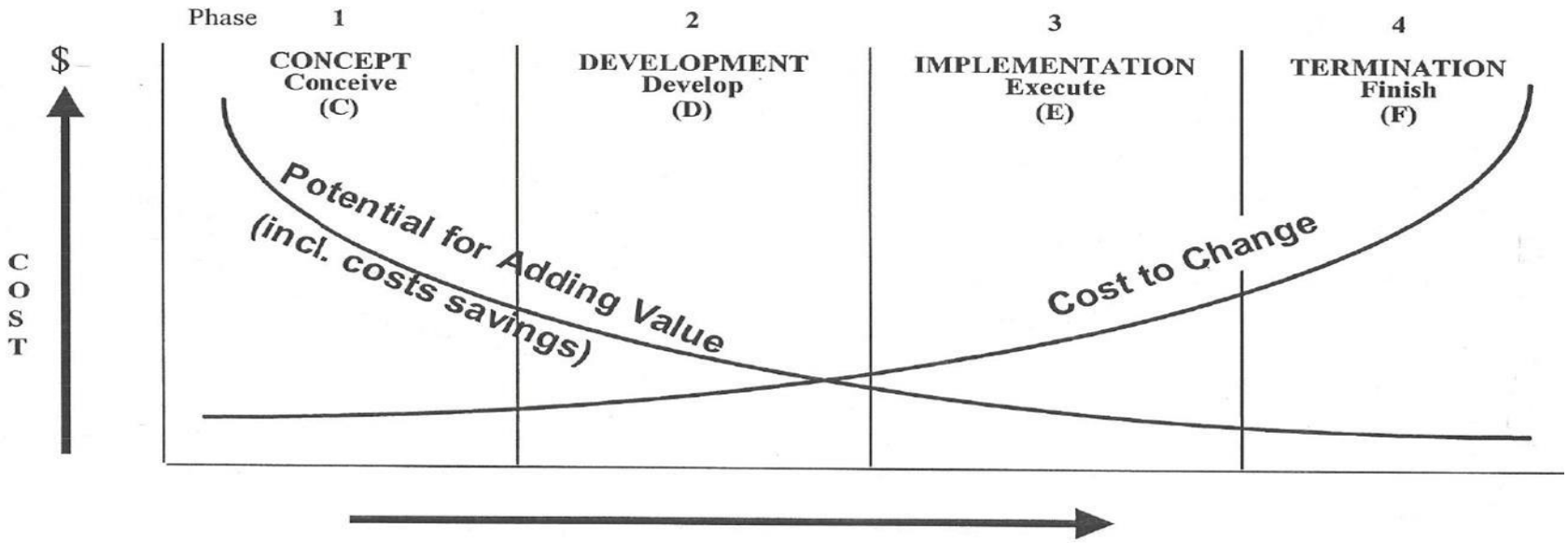
- Project Brief / Charter
- Project Manager Appointment
- Preliminary Project Plan
- Business Case
- Preliminary Project Scope Document
- Stakeholder Plan (Register & Management Strategy

Cost Influence Curve

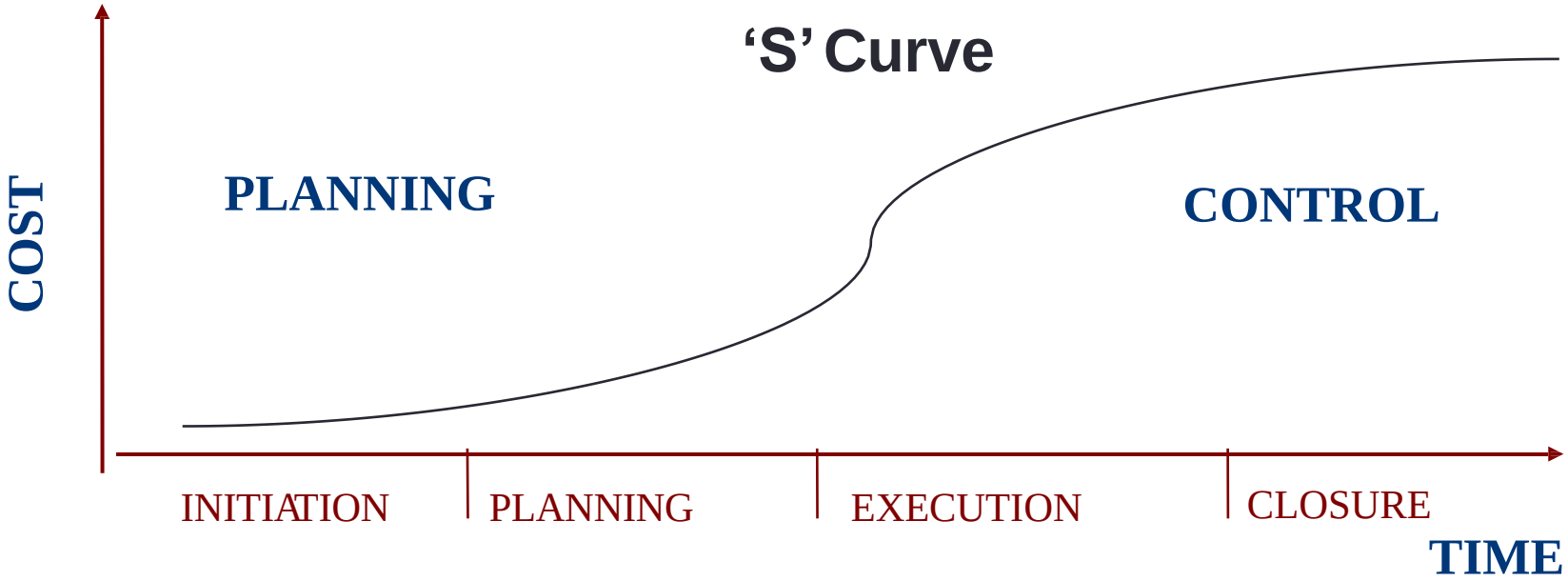
… most impact is in planning phase



Potential for:
cost savings v's cost to change



The ‘S’ curve of Project Planning



Project Planning Phase

- Quality Plan
- Update Scope Doc. & Business Case
- Preliminary Cost Estimate
- Work Breakdown Structure
- Responsibility Assignment Matrix
- Preliminary Schedule
- Communication Matrix
- Risk Management Plan
- Resource Allocation
- Business Requirements
- Specifications & Design
- Monitoring & Control
- Implementation strategy

Project Execution Phase

- Project implementation
- System procurement, manufacture, and construction
- Commission & testing
- Monitoring & control
- Implementation (Rollout)

Project Closure Phase

- Post Implementation Review
- Records Management
- Performance Reviews
- Final Sign off
- Project Closure & Handover

Project Process Groups - Activities

Review points

INITIATION

- Project Brief
- Project Manager Appointment
- Preliminary Project Plan
- Business Case
- Project Scope Document

PLANNING

- Update Scope Document & Business Case
- High Level Cost Estimate
- Work Breakdown Structure
- Responsibility Assignment Matrix
- Project Schedule
- Quality Plan
- Communication Matrix
- Risk Management Plan
- Resource Allocation
- Business Requirements
- Specifications & Design
- Monitoring & Control
- Implementation strategy
- FEL completed / signed off

EXECUTION

- Project implementation
- System procure, manufacture, and construct
- Commission & testing
- Monitoring & control
- Implementation (Rollout)

CLOSURE

- Post Implementation Review
- Records Management
- Performance Reviews
- Final Sign off
- Project Closure & Handover

Front End Loading

The **success** of a project is determined **before** sanction

'FEL concept'

Ideas

"To me, ideas are worth nothing unless executed. They are just a multiplier. Execution is worth millions." — Steve Jobs

Projects …?

Projects translate ideas into outcomes

- But is the idea worth doing???
- What are you trying to achieve?
  - What is the objective?

Starting from an idea!

- a need is identified
- new opportunity prevails
- growth path, or
- problem resolution required

What are the drivers?

Cool Home Design

Possible organisational drivers

- **Service:** improve processes
- **Productivity:** output / quality
- **Cost:** avoidance / reduction / efficiency
- **Revenue generation:** scale / opportunities
- **Customer service levels:** error rates / rework rates / customer retention rates
- **Risk:** driver may be to reduce risk

Project Goals and Constraints

- Obstacles or challenges may limit the ability to perform
  - Additional constraints
    - Amount of resources available
    - Decision maker's risk tolerance
- Opportunities may allow projects to exceed original expectations
- Which goals and constraints take precedence?

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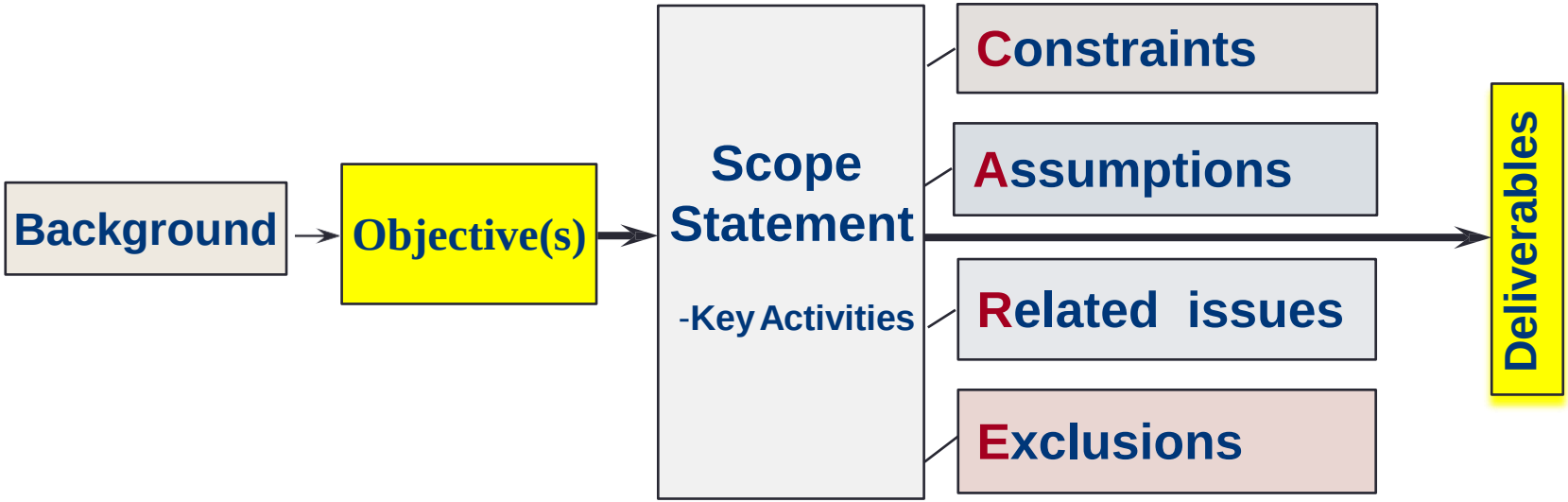
Shaping the idea into a project



Feasibility?

- How feasible is the project?
- If we do this or that …, will we get the … outcome needed?
- How do we test this?

Selecting and prioritizing Projects

What value does each potential project bring to the organization?

Are the demands of performing each project understood?

Are the resources needed to perform the project available?

Which projects will best help the organization achieve its goals?

Is there enthusiastic support both from the external customers and from one or more internal champions?

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Selecting and prioritizing Projects

- Identify potential projects
- All parts of the organization are involved
- Determine which projects align best with major goals of the firm
- Organizational priorities:
  - Understood
  - Communicated
  - Accepted

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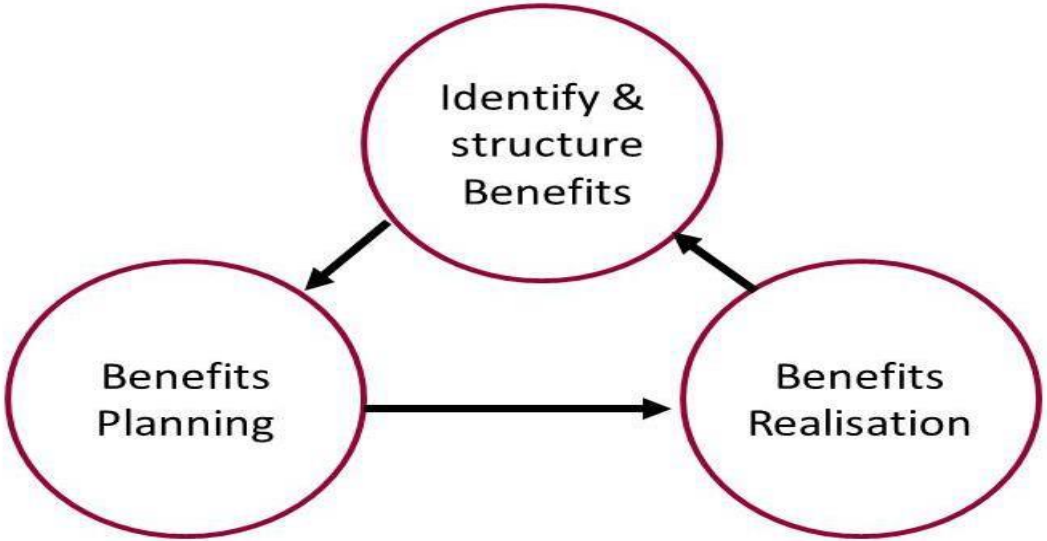
Benefits Management



Benefits Realisation - Outcomes Linkage Map

Process

Start with the end in mind

Anything we do

- **Initiative scope**
- **Contributing capability 1**
- **Contributing capability 2**

to help us transform our service operations

Change:
1. Technical
2. Organisational
3. Process
4. Service
5. Commercial

that will help translate to benefits which support our strategic goals

Interim Benefits

End Benefit

Strategic goals

Identify the benefits required to achieve strategic objectives,
Review the outcomes of the initiative that will enable the benefit to be realised

Example – Outcomes Linkage Map

Capacity building

5. Take-up of Capacity Building Program & Business Development

6. Take-up of operating subsidies

AHO Goal 8

AHO Goal 9

2021 Goals

http://www.finance.nsw.gov.au

Methods for selecting and testing the feasibility …

Which Project to proceed with!

- Methods for selecting projects
  - Focus on broad organizational needs
    - Three criteria – needs, funding, and will
      - Do people agree the project needs to be done?
      - Does the organization have the desire and capacity to provide adequate funds to perform the project?
      - Is there a strong will to make the project succeed?
  - Categorizing Projects
    - Problems, opportunities and directives
    - Time
    - Priority
  - Performing NPV or other financial analyses
    - ROI, Payback analysis

Methods for Selecting Projects

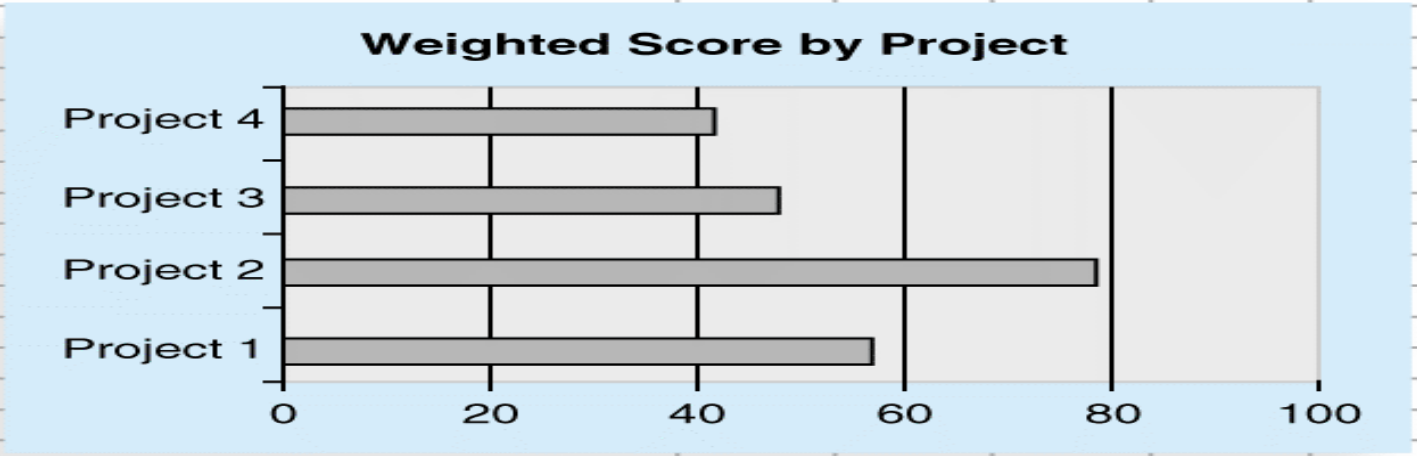
- There are usually more projects than available time and resources to implement them

- Methods for selecting projects include:
  - Using a weighted scoring model
  - Performing net present value or other financial analyses
  - Implementing a balanced scorecard

Methods for selecting projects

weighed scoring model

	A	B	C	D	E	F
1	**Criteria**	**Weight**	**Project 1**	**Project 2**	**Project 3**	**Project 4**
2	Supports key business objectives	25%	90	90	50	20
3	Has strong internal sponsor	15%	70	90	50	20
4	Has strong customer support	15%	50	90	50	20
5	Uses realistic level of technology	10%	25	90	50	70
6	Can be implemented in one year or less	5%	20	20	50	90
7	Provides positive NPV	20%	50	70	50	50
8	Has low risk in meeting scope, time, and cost goals	10%	20	50	50	90
9	**Weighted Project Scores**	**100%**	**56**	**78.5**	**50**	**41.5**



Net Present Value Analysis

- **Net present value** (NPV) analysis is a method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time
- Projects with a positive NPV should be considered if financial value is a key criterion
- The higher the NPV, the better

$$\mathrm{NPV}(i, N) = \sum_{t=0}^{N} \frac{R_t}{(1+i)^t}$$

t – the time of the cash flow

i – the discount rate,

R_t – the net cash flow

NPV

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
3	PROJECT 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
4	Benefits	$0	$2,000	$3,000	$4,000	$5,000	$14,000
5	Costs	$5,000	$1,000	$1,000	$1,000	$1,000	$9,000
6	Cash flow	($5,000)	$1,000	$2,000	$3,000	$4,000	$5,000
7	NPV →	$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	$1,000	$2,000	$4,000	$4,000	$4,000	$15,000
12	Costs	$2,000	$2,000	$2,000	$2,000	$2,000	$10,000
13	Cash flow	($1,000)	$0	$2,000	$2,000	$2,000	$5,000
14	NPV →	$3,201					
15		Formula =npv(b1,b13:f13)					
16							
17							

Note that totals are equal, but NPVs are not because of the time value of money

ROI & IRR

- **Return on investment** (ROI) is calculated by subtracting the project costs from the benefits and then dividing by the costs
  - ROI = (total benefits – total costs) / discounted costs =%
- The higher the ROI, the better

- Many organizations have an **internal rate of return (IRR)** or minimum acceptable rate of return on investment for projects
  - IRR = the discount rate at which the NPV is zero

Payback Analysis

- Another important financial consideration is payback analysis
- The **payback period** is the amount of time it will take to recoup, in the form of net cash inflows, the total dollars invested in a project
- Payback occurs when the net cumulative discounted benefits equals the costs
- Many organizations want IT projects to have a fairly short payback period

Implementing a Balanced Scorecard

- Robert Kaplan and David Norton developed this approach to help select and manage projects that align with business strategy
- **balanced scorecard**:
  - Is a methodology that converts an organization's value drivers, such as customer service, innovation, operational efficiency, and financial performance, to a series of defined metrics
- See www.balancedscorecard.org for more details

Balanced Scorecard Example

Mission: Provide responsive, professional finance and accounting services for the people who defend America

Vision:

Best Value to our customers

- World-class provider of finance and accounting services
- Trusted, innovative financial partner
- One Organization, One Identity
- Employer of choice, providing a progressive and professional work environment

Goals

- Fully satisfy customer requirements and aggressively resolve problems to deliver best value services
- Use performance metrics to drive best business practices and achieve high quality results
- Optimize the mix of our military, civilian, and contractor workforce

- Establish consultative relationships with leaders
- Deliver business intelligence to enable better decisions

- Ensure everyone is working towards the same vision and can connect what they're doing to make that vision a reality

- Embrace continuous learning for our workforce to ensure critical, high quality skill sets
- Develop the next generation of DFAS leadership

CUSTOMER PERSPECTIVE

- Improve client/customer satisfaction

FINANCIAL PERSPECTIVE

- Reduce cost to the client/customer
- Expand the use of competitive sourcing

INTERNAL PERSPECTIVE

- Improve and leverage quality
- Encourage innovation
- Deliver system solutions

GROWTH & LEARNING PERSPECTIVE

- Enhance employee competence
- Increase employee satisfication
- Enhance ability to recruit and retain DFAS talent
- Develop climate for action

eg. Combining several approaches

Criteria	**Weight**	Criterion Score			Overall value		
		P1	**P2**	**P3**	**P1**	**P2**	**P3**
Customer satisfaction	7	3	5	6		35	
Time to market	4	6	3	2		12	
Return on Investment (ROI)	4	5	6	3		24	
Net Present Value (NPV)	5	4	7	4		35	
Strategy alignment	8	7	6	7		48	
				Total Weighted Score		**154**	

Projects should never run in isolation

- Projects operate within a broad environment
- Project managers need to use **systems / strategic thinking:**
  - taking a holistic view of carrying out projects within the context of the organization
- Executives need to ensure projects continue to support current business needs

Important qualification - context

- These are all useful, age old, methods to help the path to considering feasibility but they are all limited in themselves … Meaning that they paint part of the picture but not all of the picture
- They need to be used carefully understanding the context they are being used for. Eg NPV can be a useless tool in itself without combining it with a deep understanding of stakeholder needs
- Context, context, context!

CSF's

CSF's

KPI's

KPI's

KPI's

KPI's

CSF's

CSF's

KPI's

CSF's

KPI's

What are the projects KPI's & CSF's?

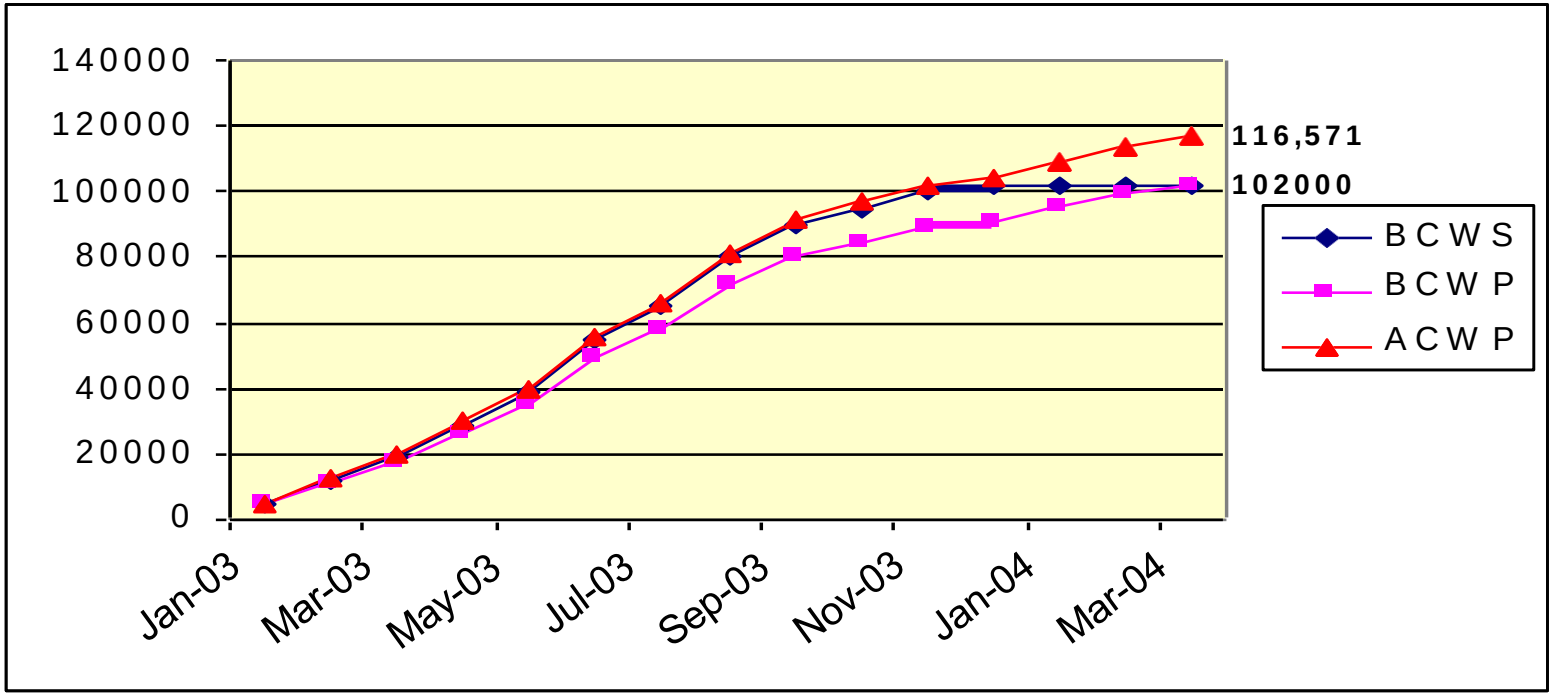
- Key Performance Indicators (KPI's)
  - Indicate whether the project is on track to achieve its objectives
  - eg could be based on cost, time, safety, etc
- They should be so aligned to Critical Success Factors such that they help achieve the CSF's
- 'What gets measured gets improved'

KPI’s should apply a SMART approach

SMART

- **S**pecific & **S**tated
- **M**easureable
- **A**greed
- **R**ealistic & **R**elevant
- **T**imely

eg KPI from Earned Value Management – Tracking a Project (cost & time)



Critical Success Factors

- The factors or elements that enable stakeholders to judge whether a Project achieved what it was meant to
  - eg Hospital patient delay times be shorter or eliminated!
- Are clear agreed business and project goals that everyone involved works to achieve
- By their nature they should be what the project gets judged on

Critical Success Factors (CSF's)

- CSF's help everyone to focus on what the Project will be judged on and therefore what they should give priority to
- CSF's need to be explicit and well communicated
- Very important to establish in the Planning phase
- CSF's are the things (normally 3 to 5) that must be achieved to ensure Project success

Summary

- BMBoK
- Ideas!
- Benefits management
- Testing feasibility
- CSF’s & KPI’s